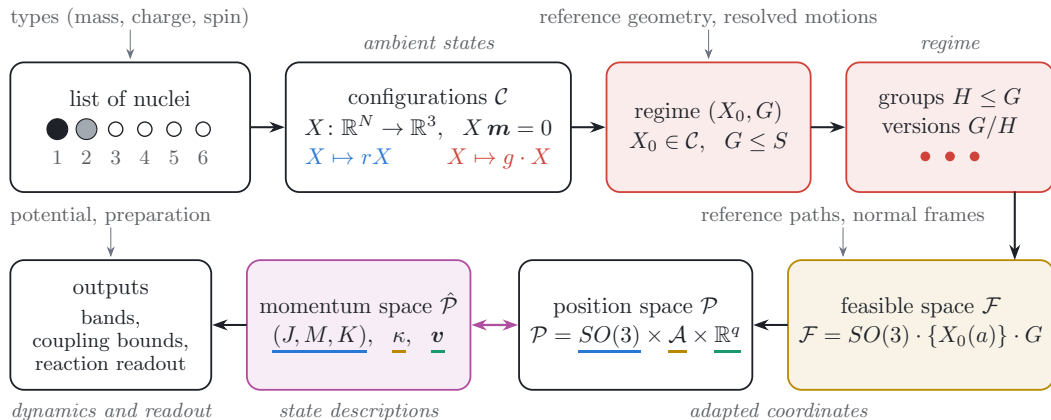


Quantum state spaces of nonrigid molecules and their complexes

Isaac AshLind

The construction



Configurations and the two actions

$$X : \mathbb{R}^N \rightarrow \mathbb{R}^3, \quad X \mathbf{m} = 0, \quad \dim \mathcal{C} = 3N - 3$$

$$r \in SO(3) : X \mapsto rX, \quad g \in S : g \cdot X = (\det g) X P_g^\top, \quad g \cdot (rX) = r(g \cdot X)$$

$$S = \left(\prod_{\text{types}} S_{n_{\text{type}}} \right) \times \{E, E^*\}$$

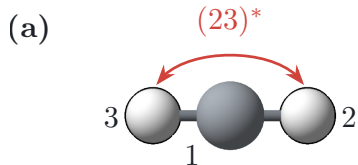
$$s = (354) \text{ acts on columns: } s \cdot X = X P_s^\top$$



States: the Pauli rule and the node rule

$$\Psi(g \cdot X) = \varepsilon(g) \chi_{\pm}(g) \sigma(g) \Psi(X) \quad (g \in S),$$

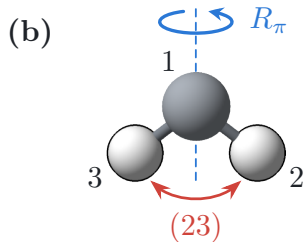
$$g \cdot X = X \Rightarrow \sigma(g) \Psi(X) = \varepsilon(g) \chi_{\pm}(g) \Psi(X)$$



$$(23)^* \cdot X = X$$

parity +: proton singlet only

parity -: proton triplet only



$$(23) \cdot X = R_{\pi} X, \text{ so } (23) \in H$$

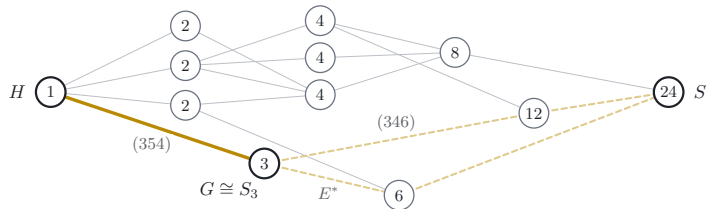
The regime, built from the bottom

$$H = \{h \in S : h \cdot X_0 \in SO(3)X_0\}, \quad R_h = (h \cdot X_0) \operatorname{diag}(\boldsymbol{m}) X_0^\top \Lambda^{-1}, \quad \Lambda = X_0 \operatorname{diag}(\boldsymbol{m}) X_0^\top$$

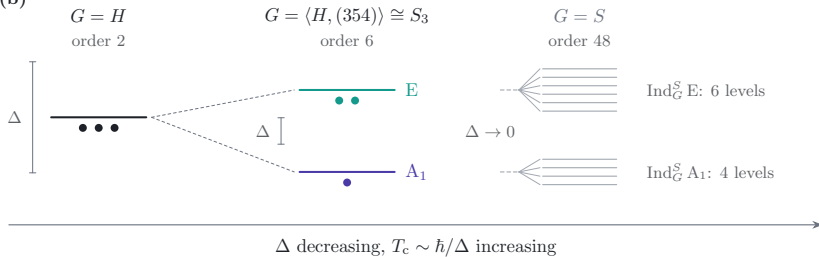
$$\text{motion } X_0 \rightarrow r(s \cdot X_0) \rightsquigarrow HsH, \quad G = \langle H, s_1, s_2, \dots \rangle \in [H, S]$$

The regime for methanol

(a)



(b)



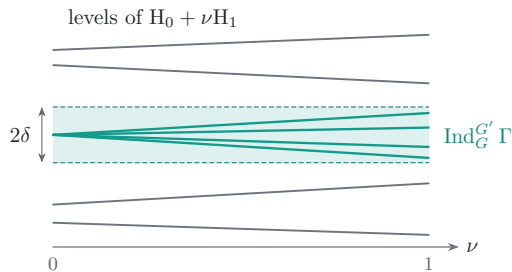
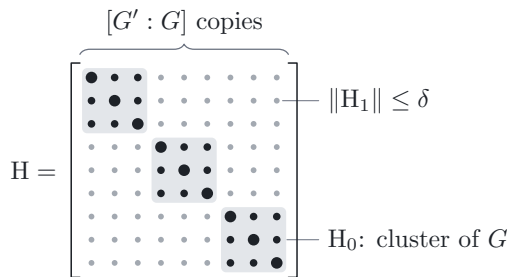
The interval as a checked search space

molecule	$ H $	turns	$ G $	bonds	$ S $	between
water	4	4	4	4	4	1
ammonia	6	6	12	12	12	2
methanol	2	6	6	12	48	2
dimethyl ether	4	36	36	144	2880	3
methylamine	2	6	12	24	240	5
hydrazine	2	2	16	16	96	8
ethane	12	36	36	144	2880	3
acetone	4	36	36	144	8640	3
PF ₅	12	12	240	240	240	4
methane	24	24	24	48	48	2
CH ₅ ⁺	2	2	240	240	240	48

Leaving couplings out

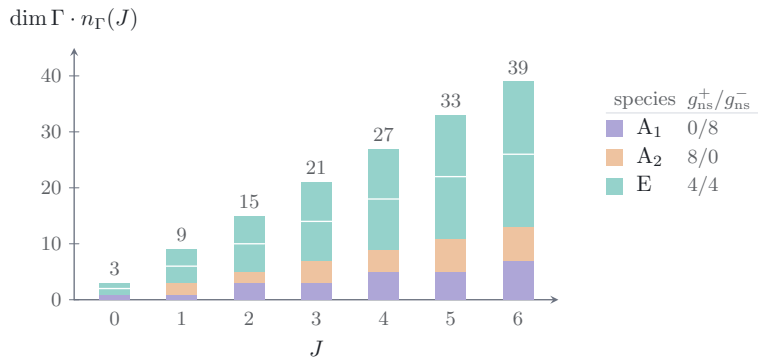
$H = H_0 + H_1$, $\|H_1\| \leq \delta$: $[G' : G]$ copies, each level of H within δ of one of H_0

$\Pr(\text{leave by } t) \leq (\delta t / \hbar)^2$, isolated by $> 2\delta$: $\Gamma \mapsto \text{Ind}_G^{G'} \Gamma$, $\delta \leq \hbar \sqrt{\gamma} / T_c$



Exact outputs: counts and spin weights

$$n_{\Gamma}(J) = \frac{1}{|H|} \sum_{h \in H} \chi_{\Gamma}(h^{-1}) \chi_J(R_h), \quad g_{\text{ns}}^{\pm}(\Gamma) = \frac{1}{|G|} \sum_{g \in G} \chi_{\Gamma}(g) \chi_{\sigma}(g) \varepsilon(g) \chi_{\pm}(g)$$

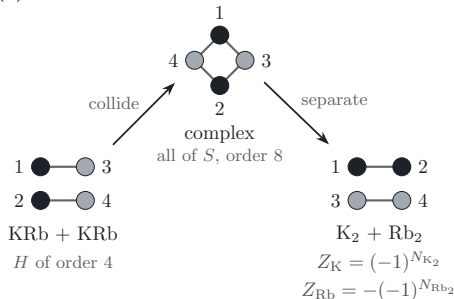


Several molecules: $\text{KRb} + \text{KRb} \rightarrow \text{K}_2 + \text{Rb}_2$

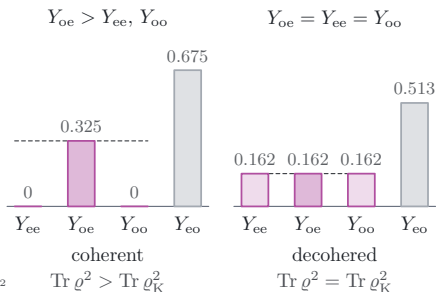
$$\Pr(Z_{\text{K}}, Z_{\text{Rb}}) = \frac{1}{4} (1 + Z_{\text{K}} \text{Tr } \varrho_{\text{K}}^2 + Z_{\text{Rb}} \text{Tr } \varrho_{\text{Rb}}^2 + Z_{\text{K}} Z_{\text{Rb}} \text{Tr } \varrho^2)$$

$$\text{separable } \varrho : \text{Tr } \varrho^2 \leq \text{Tr } \varrho_{\text{K}}^2, \text{Tr } \varrho_{\text{Rb}}^2$$

(a)



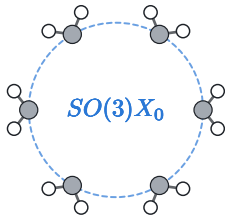
(b)



The rigid limit

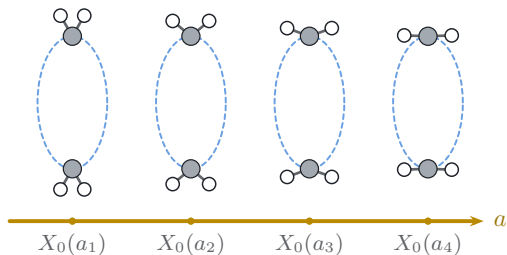
$$\mathcal{A} = \{\text{pt}\} : \quad \mathcal{F} = SO(3)X_0G \quad \text{versus} \quad \mathcal{F} = SO(3) \cdot \{X_0(a)\} \cdot G$$

(a)



one shape, every orientation

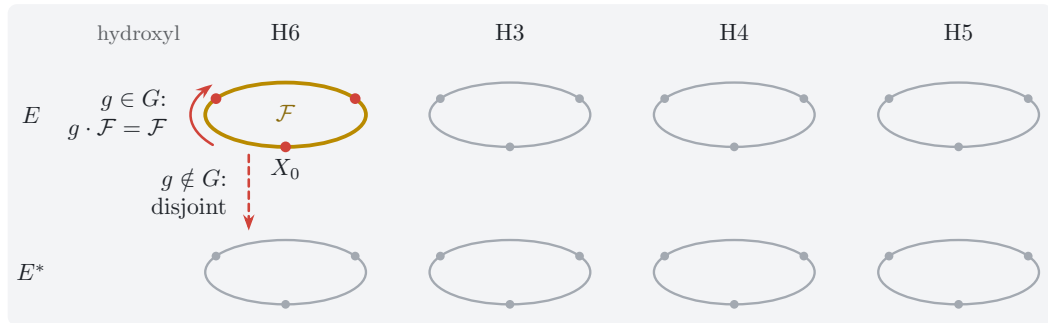
(b)



changing shapes, each with its orientations: $SO(3) \cdot \{X_0(a)\}$

The feasible configuration space

$$\mathcal{F} = SO(3) \cdot \{X_0(a)\} \cdot G, \quad g \cdot \mathcal{F} \cap \mathcal{F} = \emptyset \ (g \notin G), \quad \mathcal{A} = \mathbb{T}^{n_\tau} \times [-1, 1]^{n_\iota}$$



$\mathcal{C}$ holds $|S/G| = 48/6 = 8$ disjoint images $g \cdot \mathcal{F}$

Versions

$$|G/H| = \left(\prod_{\text{circles}} f \right) \cdot 2^{\#\text{inversions}}, \quad gH \text{ joined to } gHsH$$

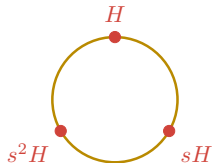
(a) water



(b) ammonia

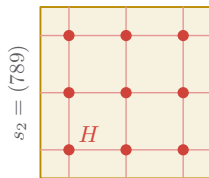


(c) methanol



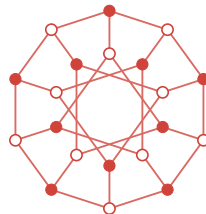
$s = (354)$

(d) dimethyl ether



$s_1 = (456)$

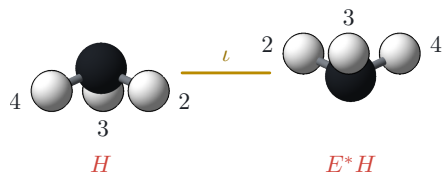
(e) PF_5



filled, open: the two hands

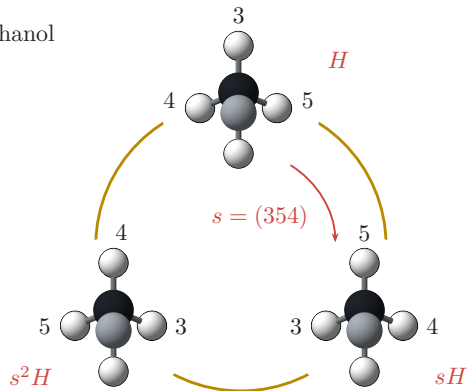
Versions as molecules

(a) ammonia



the umbrella turns inside out;
the labels 2, 3, 4 change order

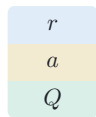
(b) methanol



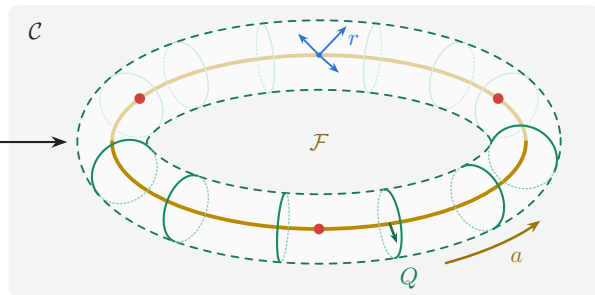
Position space

$$\mathcal{P} = SO(3) \times \mathcal{A} \times \mathbb{R}^q, \quad \varphi(r, a, Q) = r(X_0(a) + L(a)Q), \quad q = 3N - 6 - \dim \mathcal{A}$$

$$\mathcal{P} = SO(3) \times \mathcal{A} \times \mathbb{R}^q$$



$$\varphi(r, a, Q) = r(X_0(a) + L(a)Q)$$

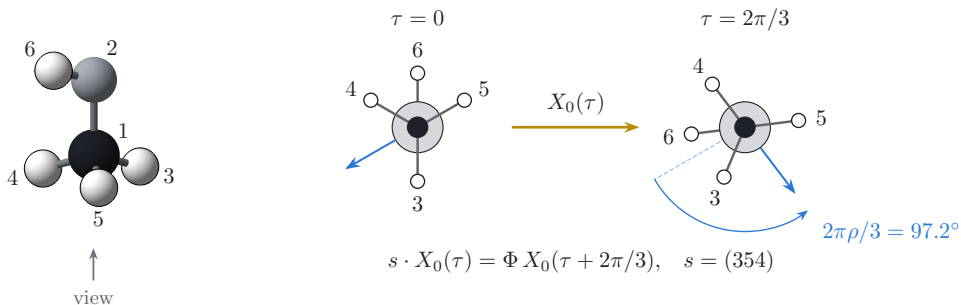


$$Q \in \mathbb{R}^q, \quad q = 3N - 6 - \dim \mathcal{A} = 11$$

The action of G on the chart

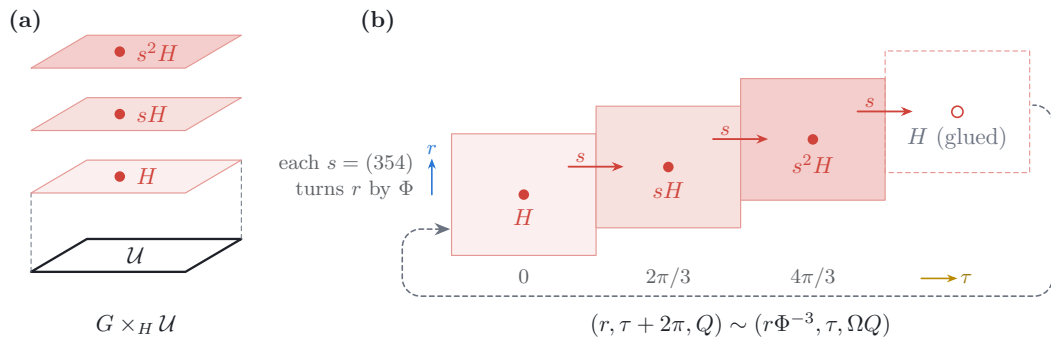
$$g : (r, a, Q) \mapsto (rR_g(a), g \cdot a, L_g(a)Q), \quad R_{gh}(a) = R_h(a) R_g(h \cdot a)$$

$$(r, \tau + 2\pi, Q) \sim (r\Phi^{-f}, \tau, \Omega(\tau)Q), \quad L(\tau)\Omega(\tau) = \Phi^f L(\tau + 2\pi), \quad \rho = I_\alpha I^{-1} \lambda$$



The atlas

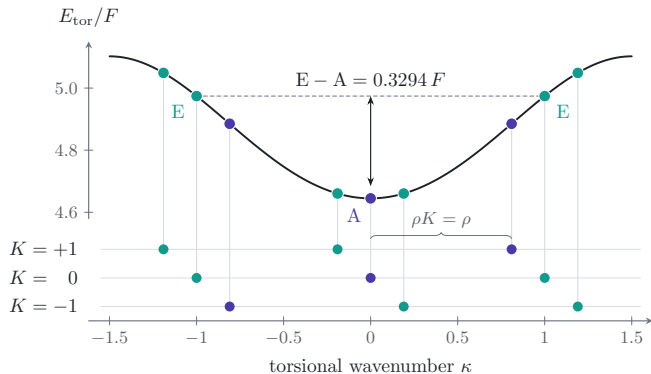
$$G \times_H \mathcal{U} = \{[g, u]\}, \quad [gh, u] = [g, h \cdot u], \quad L^2 \text{ of the region} = \text{Ind}_H^G L^2(\mathcal{U})$$



Momentum space and the torsional band

$$\hat{\mathcal{P}} = \{(J, M, K, \kappa) : \kappa \in \mathbb{Z} + \rho K\}, \quad \psi_K(\tau + 2\pi) = e^{2\pi i \rho K} \psi_K(\tau)$$

$$s e^{i\kappa\tau} |JMK\rangle = \zeta_f^k e^{i\kappa\tau} |JMK\rangle, \quad k = \kappa - \rho K \in \mathbb{Z}$$

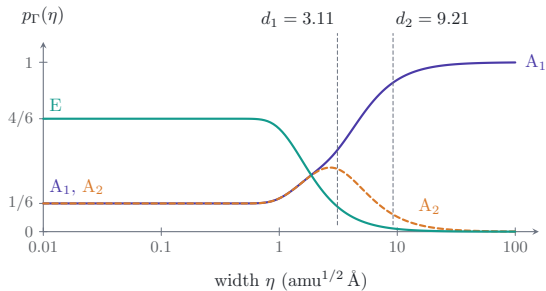


Coherent states: on the full configuration space, no chart

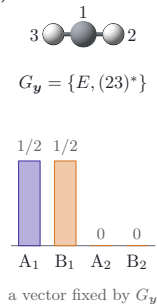
$$|\mathbf{y}, \mathbf{p}\rangle \propto e^{-|\mathbf{y}' - \mathbf{y}|^2 / 2\eta^2 + i\mathbf{p} \cdot \mathbf{y}' / \hbar}, \quad p_{\Gamma}(\eta) = \frac{\dim \Gamma}{|G|} \sum_{g \in G} \chi_{\Gamma}(g^{-1}) e^{-|\mathbf{y} - g \cdot \mathbf{y}|^2 / 4\eta^2}$$

spatial shares; physical packets also need nuclear-spin pairing and the Pauli rule

(a)



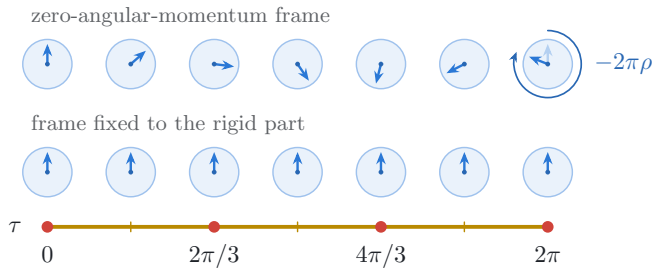
(b)



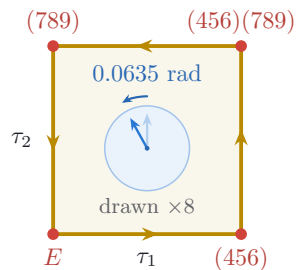
Gauge

$$r \mapsto rW(a)^{-1}, \quad R_g(a) \mapsto W(a)R_g(a)W(g \cdot a)^{-1}, \quad \psi \mapsto D^J(W)\psi$$

(a)



(b)



Key references

- ▶ Rigid orientation: Albert, Kubischta, Lemeshko and Liu, arXiv:2403.04572.
- ▶ Permutation–inversion groups: Longuet-Higgins, Mol. Phys. 6, 445 (1963); Bunker and Jensen, *Molecular Symmetry and Spectroscopy* (1998); Bunker, Natural Sciences 5, e20240036 (2025).
- ▶ Reference paths and frames: Hougen, Bunker and Johns, J. Mol. Spectrosc. 34, 136 (1970); Lin and Swalen, Rev. Mod. Phys. 31, 841 (1959); Hougen, J. Mol. Spectrosc. 114, 395 (1985).
- ▶ Mechanical connection: Guichardet, Ann. IHP A 40, 329 (1984); Shapere and Wilczek, Am. J. Phys. 57, 514 (1989); Littlejohn and Reinsch, Rev. Mod. Phys. 69, 213 (1997).
- ▶ Gaussians: Heller, J. Chem. Phys. 75, 2923 (1981); Mitroy et al., Rev. Mod. Phys. 85, 693 (2013).
- ▶ Data: Wilson, J. Chem. Phys. 3, 276 (1935); Xu et al., J. Mol. Spectrosc. 251, 305 (2008); Hu et al., Nat. Chem. 13, 435 (2021).